

## Correlation: Problem Set 2

November 10, 2025

### Question 1. Computing Correlation from Small Data

The following data show the number of hours studied ( $X$ ) and marks obtained ( $Y$ ) by 6 students:

| Student | $X$ | $Y$ |
|---------|-----|-----|
| 1       | 2   | 40  |
| 2       | 3   | 45  |
| 3       | 4   | 50  |
| 4       | 5   | 55  |
| 5       | 6   | 60  |
| 6       | 8   | 65  |

Compute the Pearson correlation coefficient between  $X$  and  $Y$ , and interpret the result.

### Question 2. Analytical Relationship between Correlation and Regression

Show that if  $b_{yx}$  and  $b_{xy}$  are the regression coefficients of  $Y$  on  $X$  and  $X$  on  $Y$  respectively, then

$$r = \text{sign}(b_{yx})\sqrt{b_{yx}b_{xy}}.$$

Also, discuss what happens to  $r$  when one of the regression coefficients is negative.

### Question 3. Effect of Linear Transformation

If  $X$  and  $Y$  are two variables, and we define

$$U = a + bX, \quad V = c + dY$$

where  $a, c$  are constants and  $b, d$  are positive constants, show that the correlation between  $U$  and  $V$  is the same as that between  $X$  and  $Y$ .

### Question 4. Interpreting Correlation in Real Context

The following table gives the average monthly temperature ( $X$ ) in  $^{\circ}\text{C}$  and electricity consumption ( $Y$ ) in units for a small town:

| Month | $X$ | $Y$ |
|-------|-----|-----|
| Jan   | 18  | 240 |
| Feb   | 21  | 260 |
| Mar   | 25  | 300 |
| Apr   | 30  | 350 |
| May   | 33  | 390 |
| Jun   | 35  | 410 |

Compute the correlation coefficient and comment whether temperature and electricity consumption are positively or negatively correlated. What real-world reasoning supports your finding?

**Question 5. Conceptual Question on Correlation vs. Causation**

The correlation between ice-cream sales and drowning incidents in a city is found to be  $+0.8$ . Explain why high correlation does not imply causation, and identify a possible lurking variable that explains the association.